

UNIT – V

Digital Cellular Systems

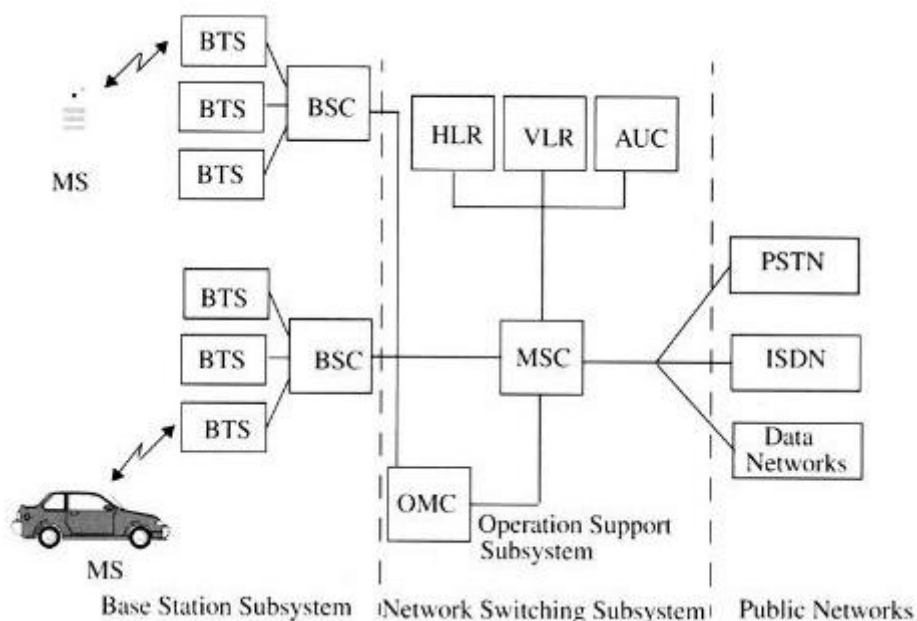
The digital cellular system has so many advantages over the analog system. The most significant advantages are digital signal processing techniques can be used to reduce transmission bandwidth requirement and to extract signals in poor SNR conditions. Digital cellular radio system uses different techniques for channel allocation as follows

1. Global system for mobile (GSM)
2. Time division multiple access (TDMA)
3. Code division multiple access (CDMA)

5.1 Global System for Mobile Systems (GSM)

The first global system for mobile (GSM) TDMA system was developed by CEPT, a European group in 1982. GSM technique is most widely used for digital cellular radio. A GSM system has maximum 200 full duplex channel per cell. Each cell has different uplink and downlink frequency. It uses digital modulation and network level architectures. GSM is a second generation (2G) cellular system. GSM can handle voice and data traffic, the voice waveform being digitally encoded before transmission. GSM transmission is done within frequency bands of 900 MHz, 1800 MHz and 1900 MHz.

GSM Architecture



GSM architecture consists of following subsystems as shown in figure 5.1

1. Mobile station (MS)
2. Base station subsystem (BSS)
3. Network and switching subsystem (NSS)
4. Operation subsystem (OSS)

Mobile Station (MS):

The mobile station consists of mobile equipment and a subscriber identity module (SIM). Mobile equipment does not need to be personally assigned to one subscriber. All the subscriber related information is stored in the SIM. When a subscriber's SIM is inserted into mobile equipment of any MS, and that MS belongs to the particular subscriber and the call is delivered to that MS. The mobile equipment is not associated with a called number, but it is linked to the SIM. The International Mobile Equipment Identity (IMEI) is assign to mobile equipment for unique identification and SIM card contain International Mobile Subscriber Identity (IMSI).

Base Station Subsystem:

The base station subsystem consists of two parts:

1. Base transceiver station (BTS)
2. Base station controller (BSC)

BSS connects to the MS through a radio interface and also connects to the network and switching subsystem (NSS). The BTS contains reception equipment and radio transmission similar to the ME in an MS. BTS contains a transcoder rate adaption unit (TRAU) that perform encoding and speech decoding and rate adaptation for transmitting data. The TRAU may be sited away from the BTS, usually at the MSC as a subpart of the BTS. BSC manages the radio resources for one or more BTS.

Network and Switching Subsystem:

The central component of network subsystem is the mobile service switching center (MSC). NSS in GSM uses an intelligent network (IN). A signaling NSS consists of main switching functions of GSM. NSS control the communication between GSM users and other telecommunications users. NSS management includes of:

- **Mobile service switching center (MSC):** MSC coordinates the call set-up to and from GSM users. Single MSC controls many BSCs.

- **Interworking function (IWF):** IWF plays the role of a gateway for MSC to interface with external networks for communication with users which are outside GSM, like circuit-switched public data network (CSPDN) or packet-switched public data network (PSPDN). The role of the IWF depends on the type network to which it interfaces and user data.
- **Home location register (HLR):** HLR consists of a stand-alone computer without switching capabilities, and maintain a database which contains subscriber information, and information related to the subscriber's current location, but not the actual location of the subscriber. A subdivision of HLR is the authentication center (AUC). The AUC manages the security data for subscriber authentication. Another sub-division of HLR is the equipment identity register (EIR) which stores the data of mobile equipment (ME) or ME-related data.
- **Visitor location register (VLR):** Links to one or more MSCs, temporarily storing subscription data currently served by its corresponding MSC, and holding more detailed data than the HLR. For example, the VLR contains more current subscriber location information than the location information at the HLR.
- **Gateway MSC (GMSC):** To set up a requested call, the call is firstly routed to a gateway MSC, and then GMSC finds the correct HLR by identifying the directory number of the GSM user. The GMSC has an interface with the external network for gatewaying the signal.
- **Signaling transfer point (STP):** STP is an essential requirement of the NSS to function itself as a stand-alone node. STP reduces the cost of the signaling transport among MSC/VLR, GMSC, and HLR.

Operation Subsystem:

Operation subsystem contains following three areas as shown in figure 5.1

1. Network operation and maintenance functions
2. Subscription management (charging and billing)
3. Mobile equipment management.

These tasks require interaction between some or all of the infrastructure equipment. OSS is implemented in any existing network.

Layer modeling (OSI Model)

The open system interconnection (OSI) of GSM consists of five layers as transmission (TX), radio resource (RR) management, mobility management (MM), communication management (CM) and operation administrator and

maintenance (OAM) as shown in figure 5.2.

- The transmission layer make the connection between MS and BTS
- The RR Layer manages the transmission over the radio interface and provides the stable link between MS and BSC.
- MM layer manages the subscriber information and authentication activity.
- The CM layer manages call control, supplementary services management and short message service(SMS)

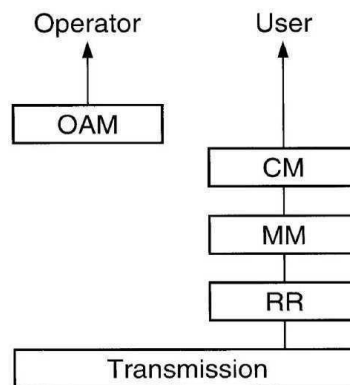


Figure 5.2 Functional planes of GSM

GSM Channel Structure:

GSM has four radio transmission modes a speech mode and three data modes for services offered to users. The radio transmission modes use the physical channels.

➤ Physical Channels:

Physical channels are used to carry either digitally encoded user speech or user data in uplink and downlink direction. There are three types of physical channels and it is also called traffic channels (TCHs):

1. TCH/F (full rate): Transmits a speech code of 13 kbps or three data-mode rates, 12, 6, and 3.6 kbps.
2. TCH/H (half rate): Transmits a speech code of 7 kbps or two data modes, 6 and 3.6 kbps.
3. TCH/8 (one-eighth rate): Used for low-rate signaling channels, common channels, and data channels.

➤ Logic channels:

Logical channels are also known as control channels. Control channels carry signaling and synchronizing commands between base station and mobile station. Types of logic channels are as follows:

1. Common channels:

All the common channels are combined in different traffic channels. All channels are grouped by the same cycle (51×8 BP), where BP is the burst period i.e. time slot. The value of BP is 577 μ s.

2. Downlink common channels:

There are five downlink unidirectional channels, shared or grouped by a TCH.

- Frequency correction channel (FCCH) repeats once every 51×8 BPs; used to identify a beacon frequency.
- Synchronization channel (SCH) follows each FCCH slot by 8 BPs.
- Broadcast control channel (BCCH) is broadcast regularly in each cell and received by all the mobile stations in the idle mode.
- Paging and access grant channel (PAGCH) is used for the incoming call received at the mobile station. The access grant channel is answered from the base station and allocates a channel during the access procedure of setting up a call.
- Call broadcast channel (CBCH): Each cell broadcasts a short message for 2s from the network to the mobile station in idle mode. Half a downlink TCH/8 is used and special CBCH design constraints exist because of the need for sending two channels (CBCH and BCCH) in parallel.

3. Uplink common channels:

The random-access channel (RACH) is the only one common uplink channel. RACH is the channel that the MS chooses to access the calls. There are two rates: RACH/F (full rate, one-time slot every 8 BP), and RACH/H (half rate, using 23-time slots in the 51×8 BP cycle, where 8 BP cycle is 4.615ms).

4. Signaling channels:

All the signaling channels have chosen one of the physical channels, and the logical channels names are based on their logical functions.

5. Slow Associated Control Channel (SACCH):

A slow-rate TCH used for signaling transport and used for non-urgent procedures, mainly handover decisions. It uses one-eighth rate. The TCH/F is always allocated with SACCH. This combined TCH and SACCH is denoted TACH/F. SACCH occupies 1-time slot (0.577 ms) in every 26 frames ($4.615\text{ms} \times 26$).

6. Fast Associated Control Channel (FACCH) :

Indicates cell establishment, authenticates subscribers, or commands a handover.

7. Stand-alone Dedicated Control Channel (SDCCH) :

Sometimes the connection between a mobile station and the network is used for passing signaling information only and not for calls. This connection may be at the user's demand or for other management operations such as updating the unit's location. SDCCH operates at a very low rate and uses a TCH/8 channel. Radio slots are assigned to users only when call penetration is required. There are two modes channel modes one is dedicated and another one idle. The use of channel mode depends on the uplink and the downlink. In GSM the uplink is the signal transmitted in the opposite direction and the downlink is the signal transmitted from the BS to the MS.

8. Voice/data channels:

Each time slot of a voice channel consists of 260 bits per block. The whole block having 316 bits. Each time slot of a data channel having 120 or 240 bits per block.

GSM Channel modes

Due to the radio spectrum limitation, individual users cannot have their own traffic control channel always. The different types of GSM channel modes are as follows

1. **Dedicated mode:** In dedicated mode traffic channels are used during call establishment and SACCH to perform location updating. TCH and SACCH are dedicated channels for both uplink and downlink channels.
2. **Idle mode:** During non-call activities, the five downlink channels are in the idle mode: FCCH; SCH; BCCH, which is regularly broadcasting. PAGCH and CBCH, which sends one message in every 2 seconds. During idle channel mode of operation the mobile station listens to the common downlink channels, and also uses SDCCH (uplink channel) to register a mobile location associated with a particular base station to the network.

Multiple Access Scheme:

GSM is a combination of TDMA and FDMA. The total 124 channels are present in FDMA, and each is 200 kHz. Both the 935-960MHz uplink and 890- 916 MHz downlink have been allocated 25 MHz, for a total of 50 MHz

duplex separation is 45 MHz. If TDMA is used within a 200-kHz channel, 8 time slots are required to form a frame, frame duration is 4.615 ms, and the time slot duration burst period is 0.577ms.

- **Frequency Hopping:** GSM using a slow frequency-hopping spread spectrum technique for radio interface. The slow hopping is defined in terms of bits per hop. Hopping rate of CDMA is 217 hops/s, if the transmission rate is 270 kbps then hopping rate will be around 1200 bits/hop. If the PAGCH and the RACH were hopping channels, then hopping sequences could be broadcast on the BCCH. The common channel is forbidden from hopping and using the same frequency.
- **Different Types of Time Slots:** In CDMA system every cell provides a reference clock, using this reference clock the time slots are defined. Each time slot is given a number (TN) which is known by the base station and the mobile station. The time slot numbering is cyclic.
- **Bursts and Training Sequences:** In TDMA, the signal transmits in form of bursts. The time interval of the burst brings the amplitude of a transmitted signal up from a starting value of 0 to its normal value. Then a packet of bits is sent by a modulated signal and after transmission of the amplitude decreases to zero. These bursts occur only at the mobile station transmission or at the base station when the adjacent burst is not transmitted. Each burst contains tail bits and training sequence. The tail bits are three 0 bits added at the beginning and at the end of each burst which provides the guard time. The training sequence is a sequence known by the receiver that trains an equalizer, a device that reduces inter-symbol interference. The training sequence bits are inserted in the middle of a time slot to mitigate inter-symbol interference

15.2 North American TDMA⁶⁻⁸

15.2.1 History

North American TDMA (NA-TDMA) is a digital cellular system sometimes called American digital cellular (ADC) or digital AMPS (DAMPS), or North American digital cellular (NADC) or IS-54 system. This TDMA system was approved and design on it was started in 1987 by a group named TR45-3 after the industry debated between frequency-division multiple access and time-division multiple access. The reason those members voted for TDMA was the big influence of European GSM, which is the TDMA system. However, the requirements of designing a digital cellular system in Europe and in North America are different. In Europe, there is a virgin band (935–960 MHz downlink and 890–915 MHz uplink) for the digital cellular system. In North America, there is no new allocated band for the digital cellular system. The digital cellular system has to share the same allocated band with the analog system (AMPS, described in Chap. 3). Also, the digital and the analog systems have to be coexistent. In this circumstance, the low-risk approach is to use the same signal signature as the analog system, i.e., FDMA. Besides, because of the urgent need for large system capacity, the time for designing a new North American system had to be very short. The North American digital system was needed to be available in 1990, in only 3 years. To design a digital FDMA system would be a straightforward task. Since the analog system is a FDMA system, all the physical data gathered for the analog system in the past 20 years could be used for designing the FDMA digital system, and design time would be shortened. On the other hand, to design a TDMA digital system in the same band shared with an FDMA analog system, much more physical data would have to be de-

veloped and time would be needed to understand them. Without a good understanding of the limitation of coexistence between two different signal signatures, FDMA and TDMA, it would be very difficult to complete a digital system with good performance in a very short time. If GSM had taken 8 years to develop, NA-TDMA might also need as much time to be revised in order for it to be mature.

Because of the requirement of coexistence, a dual-mode mobile unit was decided on; i.e., the unit can work on both analog and digital systems. In a dual-mode mobile unit, the 21 call set-up channels for the analog system are available in the unit. Why not share the same call set-up channels (analog) for both the analog voice channels and digital voice channels? In this case, no additional spectrum is needed for the digital set-up channels.⁹ The spectrum is saved for adding more digital voice. Furthermore, for the sake of speeding up the completion of North American digital systems, the call set-up channels of the digital system could be shared with the analog system to make the call processing the same between the two systems. Thus, the first phase of the NA-TDMA system could be completed earlier.

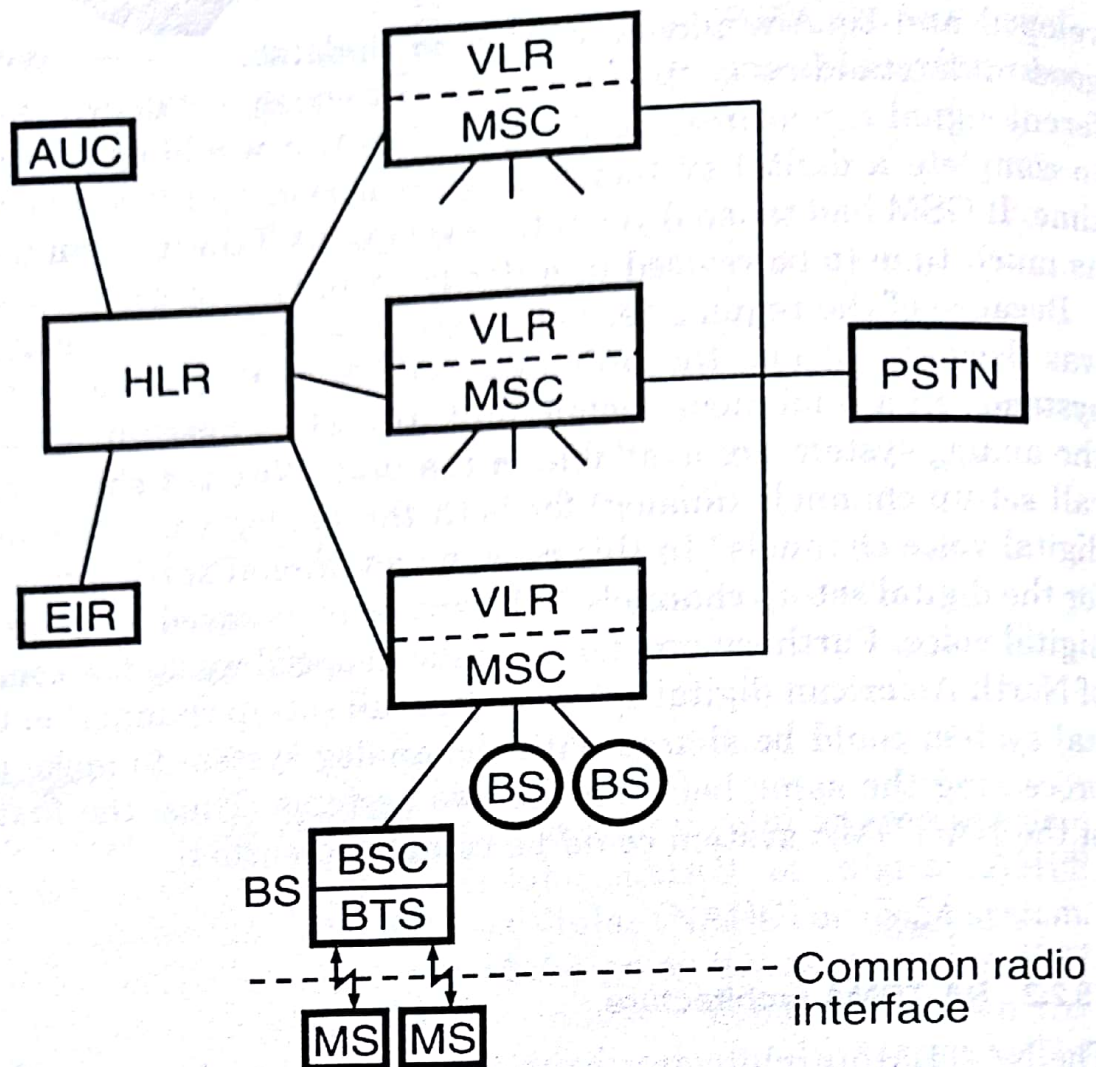
15.2.2 NA-TDMA architecture

The NA-TDMA architecture is similar to GSM architecture. The only difference is that in NA-TDMA, there is only one common interface, which is the radio interface as shown in Fig. 15.16. The NA-TDMA uses the intelligent network. All the components such as HLR, VLR, AUC, and EIR are the same as used in GSM (see Sec. 15.1.1). In developing the NA-TDMA system, there were two phases:

First phase—to commonly share the 21 set-up channels which are used for the analog system. The first-phase system is only for voice transmission. Both modes, AMPS and digital, are built in the same unit. The handoff procedure has to take care of the following four features:

1. AMPS cell to AMPS cell
2. TDMA cell to TDMA cell
3. AMPS cell to TDMA cell
4. TDMA cell to AMPS cell

Second phase—(1) generate new digital set-up channels (they were in the voice band) to access to TDMA voice channels so that a digital stand-alone unit can be provided and (2) specify a data-service signal protocol for transmitting data.



VLR: Visitor location registration
 HLR: Home location registration
 BS: Base station
 AUC: Authentication center
 EIR: Equipment identity register
 BSC: Base station controller
 BTS: Base transceiver station

Figure 15.16 NA-TDMA system architecture.

15.2.3 Transmission and modulation

TDMA structure (digital channels) In NA-TDMA, the set-up channels are analog channels shared with the AMPS system. One digital channel (a 30 kHz TDMA channel) contains 25 frames per second. Each frame is 40 ms long and has 6 time slots. Each time slot is 6.66 ms long. One frame contains 1944 bits (972 symbols), as shown in Fig. 15.17.

Each slot contains 324 bits (162 symbols) and the duration between bits is 20.57 μ s. Therefore, one radio channel is transmitted at 48.6 kbps but only 24,000 symbols per second over the radio path. Each frame consists of 6 time slots. The maximum effect on the signal for

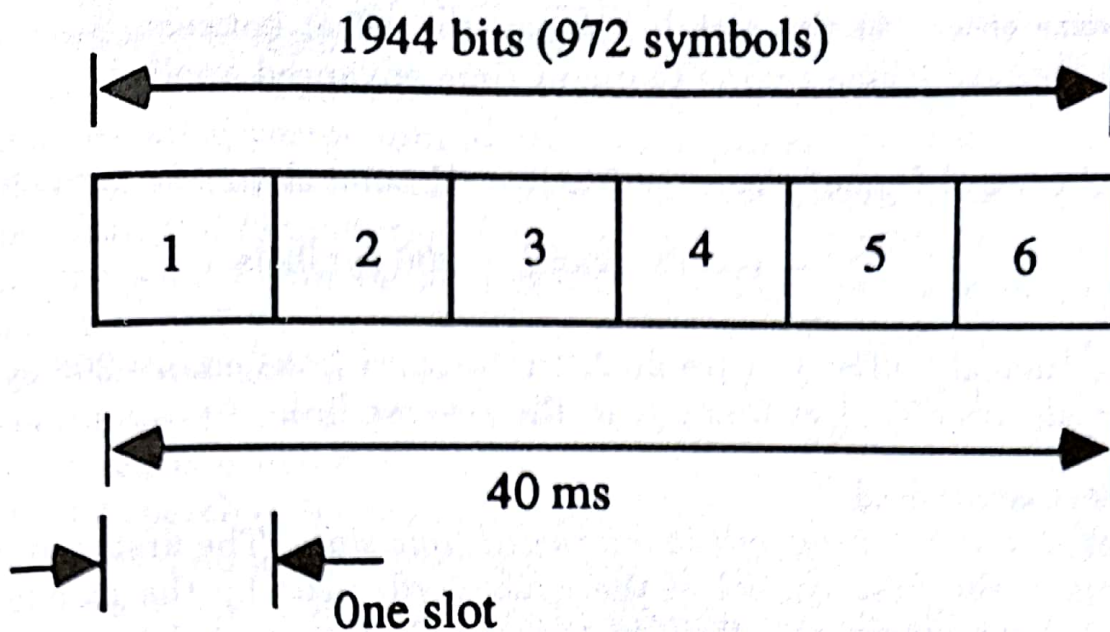


Figure 15.17 TDMA frame structure.

a forward time slot is one-half full symbol period and for a reverse time slot is 6 symbol periods (Fig. 15.18).

Frame length. There are two frame lengths, full rate and half rate. Each full-rate traffic channel shall utilize two equally spaced time slots of the frame. The overall length in each slot is shown in Fig. 15.18.

Channel 1 uses time slots 1 and 4

Channel 2 uses time slots 2 and 5

Channel 3 uses time slots 3 and 6

Each half-rate traffic channel shall utilize one time slot of the frame:

Channel 1 uses time slot 1

Channel 2 uses time slot 2

Channel 3 uses time slot 3

Channel 4 uses time slot 4

Channel 5 uses time slot 5

Channel 6 uses time slot 6

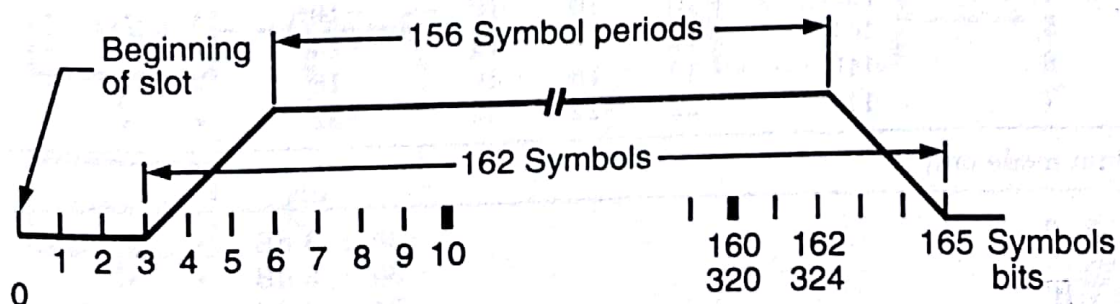


Figure 15.18 Overall length in each slot.

Frame offset. At the mobile station, the offset between the reverse and forward frame timing (without time advanced applied), is

$$\begin{aligned}\text{Forward frame} &= \text{reverse frame} + (1 \text{ time slot} + 44 \text{ symbols}) \\ &= \text{reverse frame} + 206 \text{ symbols}\end{aligned}$$

The time slot (TS) 1 of frame N (in forward link) occurs 206 symbol periods after TS 1 of frame N in the reverse link.

Modulation timing

Modulation timing within a forward time slot. The first modulated symbol (the first symbol of the sync word) used by the mobile unit shall have maximum effect on the signal (156 symbols) transmitted from the base antenna, one-half symbol (1 bit) period after beginning the time slot.

Modulation timing within a reverse time slot. The first modulated symbol has a maximum effect on the signal transmitted at the mobile unit 6 symbol periods after the beginning of the reverse time slot.

Power level. In the AMPS system, there are eight power levels. In TDMA there are an additional three levels. Therefore, in total, TDMA has 11 power levels, as shown in Table 15.2. In the carrier-off condition, the output power of the transmitting antenna must fall to -60 dBm within 2 ms. In the carrier-on condition, the output power of the transmitting antenna must come to within 3 dB of the specified level.

TABLE 15.2 Mobile Station Nominal Power Levels

Mobile station power level (PL)	Mobile attenuation code (MAC)	Nominal effective radiated power, dBW, for mobile station power class							
		I	II	III	IV	V	VI	VII	VIII
0	000	6	2	-2	-2	.	.	.	.
1	001	2	2	-2	-2	.	.	.	.
2	010	-2	-2	-2	-2	.	.	.	.
3	011	-6	-6	-6	-6	.	.	.	.
4	100	-10	-10	-10	-10	.	.	.	.
5	101	-14	-14	-14	-14	.	.	.	.
6	110	-18	-18	-18	-18	.	.	.	.
7	111	-22	-22	-22	-22	.	.	.	.
Dual-mode only									
8					-26 ± 3 dB	.	.	.	.
9					-30 ± 6 dB	.	.	.	.
10					-34 ± 9 dB	.	.	.	.

5.3 Code-Division Multiple Access (CDMA)

CDMA is the basis for many of the commercial 2G cellular systems around the world and is used in many types of communications systems like personal communication system (PCS), fixed wireless, global positioning system (GPS). CDMA is also known as IS-95. CDMA uses different codes for separation. CDMA reduces the interference by using spread-spectrum modulation. The power control scheme in a CDMA system is a requirement for digital cellular application.

Terms of CDMA Systems:

- **Active set:** The set of pilots associated with the CDMA channels containing forward traffic channels assigned to a particular mobile station (MS).
- **CDMA channel number:** An 11-bit number corresponding to the center of the CDMA frequency assignment.
- **Code channel:** Code channel is a sub channel of a forward CDMA channel. A forward CDMA channel having 64 code channels. Some code channels are assigned to different logic channels.
- **Code symbol:** It is the output of an error-correcting encoder.
- **Dim-and-burst:** A frame in which the primary traffic is multiplexed with either secondary traffic or signal traffic. It is similar to the blank-and-burst function in AMPS.
- **Forward CDMA channel:** It is a channel which is used for communication and having one or more code channels.
- **Frame:** A basic timing interval in the system. For the access channel, paging channel, and traffic channel, a frame is 20-ms long. For the sync channel, a frame is 26.666-ms long.
- **Frame offset:** A time skewing of traffic channel frames from system time in integer multiples of 1.25ms. The maximum frame offset is 18.75ms.
- **GPS (Global Position System):** System used for providing location and time information to the CDMA system.
- **Handoff (HO):** The act of transferring communication with a mobile station from one base station to another.
- **Layering:** A technique of organization of communication protocols is known as layering.
- **Long code:** A PN (pseudo noise) sequence with period using a tapped n-bit shift register.

- **Modulation symbol:** The output of the data modulator before spreading is known as modulation symbol. There are 64 modulation symbols on the reverse traffic channel. In CDMA 64-ary orthogonal modulation is used, and six code symbols are associated with one modulation symbol.

Modulation Characteristics

The reverse CDMA channel is composed of access channels and reverse traffic channels. Since the MS does not establish a system time as at the BS. The reverse channel signal received by the BS cannot use coherent detection for demodulation. So the modulation techniques for the reverse channel and forward channel are different. The modulation of the reverse channel is 64-ary orthogonal modulation at a data rate of 9600, 4800, 2400, or 1200 bps. The forward CDMA channel consists of following channels

1. Pilot Channel
2. The sync channel
3. Paging channel
4. Forward traffic channels

The actual burst transmission rate is fixed at 28,800 code symbols/second. This results in a fixed Walsh chip rate of 307.2 thousand chips per second (kcps). Every Walsh chip is spread by 4 PN chips. The rate of the spreading PN sequence is fixed at 1.2288 million chips per second (mcps).

Call Processing

Mobile station call processing in CDMA consists of following states:

- 1 MS initialization state
- 2 MS idle state
- 3 System access state
- 4 MS control on traffic channel state

➤ **MS initialization state:**

- MS selects the system to be used
- Selection of pilot channel of CDMA system within 20 ms
- Obtain timing and system configuration information of CDMA system.

➤ **MS idle state:** In MS idle state MS performs the paging channel monitoring procedures and maintain all active registration timers.

➤ **System access state:** In this state, MS sends a message to the BS on the access channel, and from BS it receives messages through paging

channel this process called access attempt.

- **MS control on traffic channel state:** MS communicate with BS through reverse and forward traffic channel and perform following functions
- MS verifies the transmission and reception of traffic channels.
 - Wait for an order information message with alert
 - Wait for the answer of call by user.
 - After the completion of communication disconnects the call.

Handoff Procedures

The MS continuously searches for pilots to detect the availability of other CDMA signals that are on the same carrier frequency and measures or calculate the pilot strength. When the mobile station detects a pilot of sufficient strength which is not associated with the serving cell. Then MS sends a message to the serving base station. The cellular network decides which neighbor base stations can be involved in a handoff and selects an idle Walsh function associated with the selected site, effectively selecting a traffic channel. Types of handoff of CDMA are as follows:

➤ **Soft Handoff**

The condition where two cells are in simultaneous communication with the mobile is called Soft Handoff. MS make communication with new BS without disturbing communication with old BS. Same frequency is assigned in soft handoff between old and new BS.

➤ **CDMA to CDMA hard handoff**

Different frequency assignment is done for two BS for the transmission by MS.

➤ **Softer Handoff**

Softer Handoffs entail the simultaneous serving of a mobile unit by two sectors within same cell.

➤ **CDMA to Analog Handoff**

If no CDMA channel is available for handoff, then the call would be handed off to an available analog channel at the serving base station and switched to the analog mode of processing. MS directed to an analog channel with different frequency.

5.4 Examples for 2G, 3G and 4G systems

Mobile communication is one of the hottest areas with advanced techniques. It is developing extremely fast in present times and deals with all the fields of mobile and wireless communications. The evolution and development of various generations of mobile wireless technology along with their advantages and disadvantages are discussed below.

1G

- This is the first generation of wireless telephone technology, mobile telecommunications, which was launched in Japan by NTT in 1979.
- The main technological development in this generation that distinguished the First Generation mobile phones from the previous generation was the use of multiple cell sites, and the ability to transfer calls from one site to the next site as the user travelled between cells during a conversation.
- It used analog signals.
- Its Speed was up to 2.4kbps.
- It allowed only voice calls and in one country.
- Not possible to send or receive text messages.

Disadvantages

- Poor quality of voice
- Poor life of Battery
- Size of phone was very large
- No security
- Capacity was limited
- Poor handoff reliability

2G

- This is the second generation of mobile telecommunication was launched in Finland in 1991.
- It was based on GSM standard.
- It used digital signals.
- It enables data transmission like as text messaging (SMS - Short Message Service), transfer of photos or pictures (MMS), but not videos.
- The later versions of this generation, which were called 2.5G

- using GPRS (General Packet Radio Service) and 2.75G using
- EDGE (Enhanced data rates for GSM Evolution) networks.
- It provides better quality and capacity.

Disadvantages

- Unable to handle complex data such as Video
- Requires strong digital signals

3G

- 3G is the third generation was introduced in early 2000s.
- The transmission of data was increased up to 2Mbits/s, which allows you to sending or receiving large email messages.
- The main difference between 3G and 2G is the use of packet switching rather than circuit switching for data transmission.
- Faster communication
- High speed web or more security
- Video conferencing
- 3D gaming
- TV streaming, Mobile TV, phone calls etc. are the features of 3G.

Disadvantages

- Costly
- Requirement of high bandwidth
- Expensive 3G phones
- Size of cell phones was very large.

4G

- 4G is the fourth generation of mobile telecommunication which was appeared in 2010.
- It was based on LTE (Long Term Evolution) and LTE advanced standards.
- Offer a range of communication services like video calling, real time language translation and video voice mail.
- It was capable of providing 100 Mbps to 1Gbps speed.
- High QoS (Quality of Service) and High security.
- The basic term used to describe 4G technology is MAGIC. Where :
 - M - Mobile multimedia

- A - Anytime anywhere
- G - Global mobility support
- I - Integrated wireless solution
- C - Customized personal service

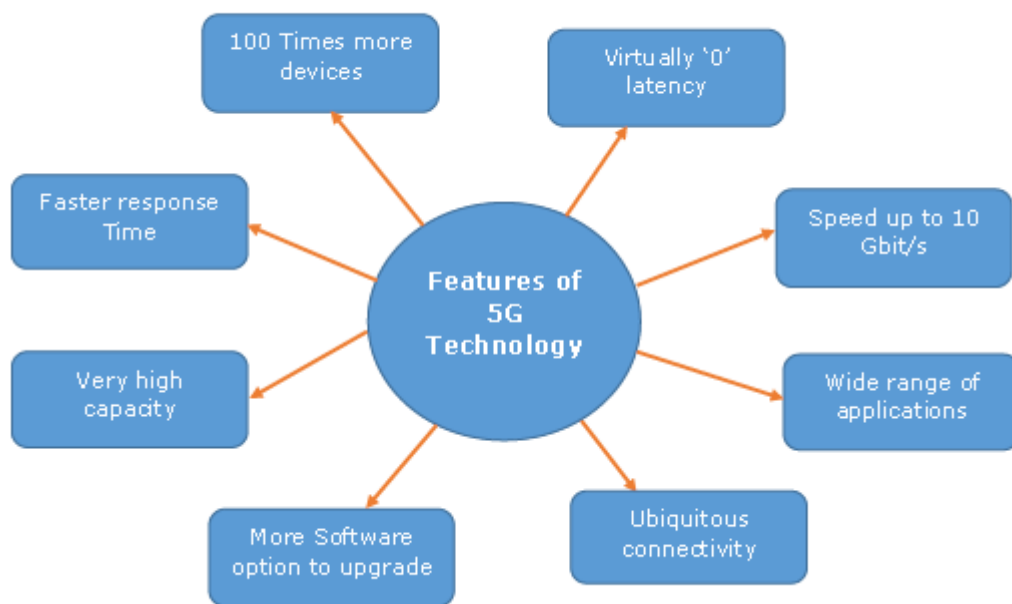
Disadvantages

- Uses more battery
- Difficult to implement
- Expensive equipment are required

5.5 Introduction to 5G system

Salient Features of 5G

5th Generation Mobile Network or simply 5G is the forthcoming revolution of mobile technology. The features and its usability are much beyond the expectation of a normal human being. With its ultra-high speed, it is potential enough to change the meaning of a cell phone usability.



With a huge array of innovative features, now your smart phone would be more parallel to the laptop. You can use broadband internet connection; other significant features that fascinate people are more gaming options, wider multimedia options, connectivity everywhere, zero latency, faster response time, and high quality sound and HD video can be transferred on other cell phone without compromising with the quality of audio and video.

What is 5G Technology?

The 5G technology is expected to provide a new (much wider than the previous one) frequency bands along with the wider spectral bandwidth per frequency channel. As of now, the predecessors (generations) mobile technologies have evidenced substantial increase in peak bitrate. Then – how is 5G different from the previous one (especially 4G)? The answer is – it is not only the increase in bitrate made 5G distinct from the 4G, but rather 5G is also advanced in terms of –

- High increased peak bit rate
- Larger data volume per unit area (i.e. high system spectral efficiency)
- High capacity to allow more devices connectivity concurrently and instantaneously
- Lower battery consumption
- Better connectivity irrespective of the geographic region, in which you are Larger number of supporting devices
- Lower cost of infrastructural development
- Higher reliability of the communications

As researchers say, with the wide range of bandwidth radio channels, it is able to support the speed up to 10 Gbps, the 5G *WiFi* technology will offer contiguous and consistent coverage – “wider area mobility in true sense.”

Application and Uses of 5g Technology

- Mobile Network With High Speed
- Multimedia and Entertainment
- Internet of Things
- Smart Cities
- Health Industry
- Autonomous Cars

Advantages and limitations of 5G Technology

Advantages of 5G Technology:

1. **Increased Speed:** 5G is 100 times faster than the current 4G networks, enabling users to download files, stream content, and interact with others quickly and efficiently.
2. **Lower Latency:** Latency is the time it takes for a signal to travel from one point to another. 5G has a very low latency rate, meaning the response time between devices is lowered.

3. **Improved Connectivity:** 5G can support more connected devices than 4G.

Disadvantages of 5G Technology:

1. **Cost:** 5G technology is still in its infancy, and the cost of implementation is high. This could prevent some businesses from utilising the technology.
2. **Security:** 5G networks are still vulnerable to cyber-attacks, so businesses must ensure their networks are secure.
3. **Accessibility:** 5G technology is unavailable in many areas, and the coverage is still limited. This can limit the ability of businesses to take advantage of the technology.

Comparison	2G	3G	4G	5G
Introduced in year	1993	2001	2009	2018
Technology	GSM	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	TDMA, CDMA	CDMA	CDMA	OFDM, BDMA
Switching type	Circuit switching for voice and packet switching for data	Packet switching except for air interference	Packet switching	Packet switching
Internet service	Narrowband	Broadband	Ultra broadband	Wireless World Wide Web
Bandwidth	25 MHz	25 MHz	100 MHz	30 GHz to 300 GHz
Advantage	Multimedia features (SMS, MMS), internet access and SIM introduced	High security, international roaming	Speed, high speed handoffs, global mobility	Extremely high speeds, low latency
Applications	Voice calls, short messages	Video conferencing, mobile TV, GPS	High speed applications, mobile TV, wearable devices	High resolution video streaming, remote control of vehicles, robots, and medical procedures